

CONTENT

NORMATIVE REFERENCES.....	—
DESIGNATIONS AND ABBREVIATIONS.....	—
DEFINITIONS	—
INTRODUCTION	—
1 PROBLEM DOMAIN ANALYSIS AND CURRENT STATE OF AUTOMATED DIABETIC RETINOPATHY DIAGNOSIS	1
1.1 Medical and Epidemiological Context of Diabetic Retinopathy	1
1.1.1 Pathophysiology and Clinical Grading Systems	1
1.1.2 Screening Requirements in Resource-Limited Healthcare Settings	4
1.2 Fundus Image Acquisition and Quality Variability	6
1.2.1 Sources of Image Degradation in Clinical Practice	6
1.2.2 Impact of Image Quality on Diagnostic Model Performance	8
1.2.3 Device-Specific Variability in Fundus Imaging.....	10
1.3 Deep Learning Approaches to Retinal Image Classification.....	13
1.3.1 Convolutional Neural Network Architectures for Medical Imaging	13
1.3.2 Transfer Learning and Self-Supervised Pretraining in Ophthalmic Diagnostics	15
1.3.3 Explainability Methods in Medical Image Classification.....	17
1.4 Critical Analysis of Existing Automated DR Screening Systems	19
1.5 Formulation of the Research Problem and Justification of Research Direction...	24
Conclusions to Chapter 1.....	25
2 THEORETICAL FOUNDATIONS OF IMAGE PREPROCESSING AND DEEP LEARNING FOR FUNDUS IMAGE ANALYSIS	27
2.1 Mathematical Foundations of Image Enhancement Techniques	27
2.1.1 Histogram Equalization and Adaptive Contrast Enhancement.....	27
2.1.2 Formalization of CLAHE with Dual-Constraint Clip Limit.....	30
2.1.3 Spatial Filtering and Noise Reduction Methods	34
2.2 Theoretical Framework of Convolutional Neural Networks	36
2.2.1 Convolution, Pooling, and Feature Extraction Operations	36
2.2.2 Loss Functions and Optimization for Imbalanced Medical Datasets	38
2.2.3 Regularization Techniques: Dropout, Batch Normalization, and Data Augmentation	41
2.3 Transfer Learning and Self-Supervised Representation Learning Theory	42
2.3.1 Feature Transferability Across Visual Domains	42
2.3.2 Frozen-Layer versus Progressive Fine-Tuning Strategies	44
2.3.3 In-Domain Self-Supervised Pretraining for Retinal Imaging	46
2.4 Mathematical Modeling of Laser-Tissue Interaction in Retinal Therapy.....	48
2.4.1 Coupled Thermal-Optical Model of Fundus Tissue Response (incl. implications for diagnostic image feature interpretation)	48

2.5 Explainability in Deep Learning for Medical Imaging.....	50
2.5.1 Class Activation Mapping (CAM) and Grad-CAM Mathematical Formulation	50
2.5.2 Interpretation of Attention Maps in Ophthalmic Context	52
2.5.3 ALO and IoU as Quantitative Explainability Metrics.....	54
2.6 Image Quality Metrics for Preprocessing Evaluation	55
Conclusions to Chapter 2.....	58
3 METHODOLOGY OF INTEGRATED PREPROCESSING-CNN PIPELINE DESIGN.....	59
3.1 Formalization of the Unified Preprocessing Pipeline	59
3.1.1 Pipeline Stage Specification: 8-Stage System.....	59
3.1.2 Modified CLAHE Algorithm with Dual-Constraint Clip Limit	75
3.1.3 Augmentation Strategy for Class Imbalance Mitigation.....	77
3.1.4 External Image Ingestion Protocol	78
3.2 Design of CNN Architectures for DR Classification.....	80
3.2.1 ResNet-50 and EfficientNet-B3 as Primary Experimental Architectures....	80
3.2.2 Historical Reference Architectures (Reference Only)	83
3.3 Transfer Learning and Pretraining Methodology.....	84
3.3.1 Architecture Adaptation for Five-Class DR Classification.....	84
3.3.2 Ophthalmology-Specific Self-Supervised Pretraining of the CNN Backbone (Integrated Arm)	85
3.3.3 Two-Stage Fine-Tuning Protocol Design.....	87
3.3.4 Weighted Loss Function Formulation for Ordinal Class Structure	88
3.4 Evaluation Framework and Performance Metrics	90
3.4.1 Multi-Metric Assessment Framework.....	90
3.4.2 Cross-Validation and Statistical Reliability Protocols.....	94
Conclusions to Chapter 3.....	96
4 EXPERIMENTAL RESEARCH — PREPROCESSING IMPACT ON CNN DIAGNOSTIC PERFORMANCE	97
4.1 Datasets and Experimental Configuration	97
4.1.1 Tiered Dataset Architecture.....	97
4.1.2 Class Distribution Analysis and Data Partitioning Strategy	103
4.1.3 Hardware Specification and Reproducibility Protocol.....	105
4.2 Experiment 1: Integrated Pipeline Dominance — Pipeline + In-Domain Pretraining vs. Baseline on EyePACS (H-1)	—
4.2.1 Restored 2×2 Factorial Design (Configurations A–D)	—
4.2.2 Training Dynamics and Convergence Analysis	—
4.2.3 Quantitative Comparison of Diagnostic Metrics.....	—
4.3 Experiment 2: Pipeline Stage Ablation + CLAHE/ σ Sweeps (H-2).....	—
4.3.1 Ablation Design (Levels 0–6)	—

4.3.2 CLAHE Threshold Sensitivity Analysis (H-2 Sub-Analysis).....	—
4.3.3 Flat-Field σ Sweep and Image Quality Metrics	—
4.4 Experiment 3: Cross-Dataset Transferability on APTOS 2019 (H-4).....	—
4.4.1 Zero-Shot Transfer to APTOS 2019	—
4.4.2 Baseline vs Pipeline Comparison	—
4.5 Experiment 4: Grad-CAM Explainability on IDRiD + Clinical (H-5).....	—
4.5.1 Grad-CAM Generation Protocol	—
4.5.2 Quantitative ALO and IoU with IDRiD Lesion Masks	—
4.5.3 Attention Consistency Across Datasets.....	—
4.6 Experiment 5: Clinical Degradation Resistance on IDRiD + Messidor-2 (H-7)..	—
4.7 Experiment 6: Device Domain Shift on DDR + ODIR-5K + RFMiD (H-6)	—
4.8 Experiment 7: Small Data Training (IDRiD $\rightarrow$ Clinical).....	—
Conclusions to Chapter 4.....	—
5 RELIABILITY VALIDATION AND COMPARATIVE ANALYSIS	—
5.1 Explainability Results	—
5.2 Statistical Validation	—
5.2.1 Bootstrap Confidence Intervals and Mixed-Effects Model	—
5.2.2 Final Claim Strength Classifications	—
5.3 Comparative Analysis with Published Systems.....	—
5.3.1 Benchmarking Against Published Results: IDx-DR, EyeNuk, DeepMind..	—
5.3.2 Performance-Complexity Trade-Off Analysis	—
5.4 Limitations and Boundary Conditions of the Proposed Approach	—
Conclusions to Chapter 5.....	—
6 ARCHITECTURE OF AN AUTOMATED DR SCREENING SYSTEM FOR RESOURCE-LIMITED ENVIRONMENTS	107
6.1 System Requirements and Design Principles.....	107
6.1.1 Functional and Non-Functional Requirements Specification	107
6.1.2 Modular Architecture with PACS and EHR Integration.....	110
6.2 AI Processing Module Design	113
6.2.1 Preprocessing Engine with Configurable Pipeline Parameters.....	113
6.2.2 Inference Module with Model Selection Logic.....	115
6.3 Clinical Workflow Integration	117
6.3.1 Telemedicine and Portable Device Support for Rural Deployment.....	117
6.3.1.1 Deployment in distributed telemedicine systems	117
6.3.1.2 Integration with national eHealth platforms	118
6.3.1.3 Real-time remote DR screening in low-resource regions.....	119
6.3.2 Physician-in-the-Loop Decision Support Interface.....	120
6.4 Data Security and Regulatory Compliance Framework	121
6.4.1 GDPR/HIPAA-Aligned Data Management Protocols	121
6.4.2 Applicability to Kazakhstan Healthcare Infrastructure.....	122

Conclusions to Chapter 6.....	124
CONCLUSION	—
REFERENCES	—
APPENDIX A	—
APPENDIX B	—
APPENDIX C	—
APPENDIX D	—
APPENDIX E	—
APPENDIX F	—